Jacob Rome

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Aerospace Engineering; Cumulative GPA: 3.93/4.00 — *Aug. 2024 – Dec. 2027*

University of Minnesota — Minneapolis, MN
Dual Enrollment; Cumulative GPA: 4.00/4.00 — *Sept. 2023 – May. 2024*

Experience

NASA RAVEN Project — Jun. 2026 – Current
Research Assistant — *Atlanta, GA*

- Engineered an in-house CNC and fiber laser PCB workflow achieving 0.1mm precision on 2 layer FR4 copper boards
- Cut PCB turnaround from industry-standard days/weeks to 1-2 hours in-house, enabling faster design-test-iterate cycles
- Optimized a MATLAB script to test liquid cooling performance for motors on a 500V, 25-40A eVTOL surrogate vehicle
- Reduced required pump voltage 62%(24V→9V, 4.6→1.6 LPM), enabling a smaller pump/battery and lighter aircraft

Drone, Radio Control, Experimental Aircraft Club (DRXC) — Aug. 2025 – Current
Chief Engineer, Winner of the 2026 VFS DBVF Competition — *Atlanta, GA*

- Led system-level UAV design as Chief Engineer, coordinating 6 subteams through weekly technical design reviews
- Designed and CNC-manufactured carbon fiber airframe components, performing FEA to validate structural performance
- Reduced airframe mass by 20% using SolidWorks topology optimization, FEA, and lightweight structural redesign
- Designed a computer vision-guided linear rail pickup mechanism to autonomously secure a 3-lb payload during flight

UpperStory — Summer 2024 & Summer 2025
Research and Development Intern — *St. Paul, MN*

- Designed from-scratch charger PCB and battery interface PCB enabling modular dual-battery power & charging system
- Validated battery terminal interface under 800 mA continuous load to ensure reliable electrical contact design
- Added controller functionality by redesigning PCB inputs with a button matrix and integrating a bumper system
- Engineered a 4WD wind-up car drivetrain, improving uphill performance using design iteration and resin 3D printing

NASA L'SPACE NPWEE — Jan. 2025 – May. 2025
Project Manager and Primary Reviewer — *Atlanta, GA*

- Led team developing NASA NPWEE proposal on spacecraft debris protection, managing risk, budget, and schedule
- Developed NASA proposal using requirements-driven design informed by feedback from NASA engineers and mentors
- Chaired formal proposal review, evaluating another team's design against NASA NPWEE scoring criteria and rubric

Yellow Jacket Space Program — Jan. 2025 – Aug. 2025
Darcy Rocket Structures Team — *Atlanta, GA*

- Engineered and designed a mold for the Darcy nose cone, which was optimized for manufacturing constraints
- 3D printed the molds with ABS and used acetone surface treatment to create a smooth finish for the layup process

FIRST Tech Challenge Robotics — Sept. 2019 – Feb. 2024
Team Captain — *Woodbury, MN*

- Led the team to appear in 3 consecutive state championship tournaments, winning awards in each appearance
- Directed design and manufacturing sub-teams, introducing new systems to produce 160+ 3D printed parts
- Received the Dean's List Finalist award and personally represented Minnesota at the World Championship tournament

Projects

- **3D Printed Clock**: Engineered and assembled a fully 3D printed clock with a torsional energy system and escapement mechanism, modeling components in CAD and iterating through prototypes to achieve reliable timekeeping
- **Tensile Test Machine**: Designed and built a tensile test machine using a load cell and Arduino to measure stress-strain behavior of multiple materials, identifying of material properties through data acquisition and analysis
- **Rubik's Cube Solver**: Developed a Rubik's Cube solving robot using a color sensor for initial state detection, stepper motors for precise actuation, and algorithmic solution planning to autonomously solve the cube
- **Portfolio Website**: Designed and deployed an interactive portfolio website using React and JavaScript to showcase engineering projects, technical experiences, and CAD models in a structured, user friendly format

Skills

Technical: CAD, CAM, FEA, Electrical Schematics, PCB Design, GD&T, Fluid Dynamics, Control Systems
Tools: 3D Printing (FDM, Resin), Laser Cutting (CO2, Fiber), CNC Mill, MCUs, PCB Assembly, Transducers, Soldering
Programming: MATLAB, Simulink, C/C++, React, JavaScript, HTML
Software: Fusion 360, SolidWorks, Inventor, NX, ArduPilot, ANSYS, Arduino IDE, VS Code